

Development of a Compact Slotted Patch Antenna for Early Stage Breast Tumor Detection Using Microwave Technology

¹Sayed Shifat Ahmed, ²Ajay Krishno Sarkar

^{1,2}Department of Electrical & Electronic Engineering, Rajshahi University of Engineering and Technology
E-mail: ¹shikor99@gmail.com, ²sarkarajay139@gmail.com

Abstract— This research develops a compact slotted microstrip patch antenna that can be used in conjunction with microwave imaging techniques for early detection of breast tumors. It was developed on a low-cost FR-4 substrate and is specifically designed to work at an operational frequency of 1.9612 GHz; as such it will provide a high level of reflection and return loss (-41.27 dB) thereby minimizing loss in signal radiating from the 1.94-1.99 GHz frequency band. At lower frequency levels, the narrow bandwidth provides enhanced penetration capability allowing for better discrimination between normal and malignant breast tissue due to the higher dielectric contrast. In addition, SAR analysis confirmed maximum value of 0.623 W/kg is significantly less than the international exposure limits. Using a multi-layer breast phantom model, the simulation results demonstrate the feasibility of utilizing this antenna as part of a non-invasive, real time diagnostic system for breast cancer detection and localization analysis through variations of the scattering parameters by independent excitation of each element. Overall, this miniaturized antenna has demonstrated its potential for integration into diagnostic systems for the detection and monitoring of breast cancer through non-invasive and safe methods.

Keywords—Breast cancer, microwave imaging, tumor detection, breast phantom, tumor localization, SAR, slotted patch antenna.

I. INTRODUCTION

Breast tumor detection is being made possible through wireless wearable devices by new and innovative techniques. Due to the costs associated with conventional imaging (MRI/Mammography) and the use of ionizing radiation, there are limitations to these technologies. The increasing incidence and mortality rates from cancer globally, particularly breast cancer in women, continue to be driven by the rapid expansion of the world's population and growing numbers of older individuals who adopt western lifestyle behaviors that increase an individual's risk of developing cancer [1]. By 2040, newly diagnosed breast cancer cases are projected to rise by over 40% to about 3 million annually, while deaths are expected to increase by more than 50%, from 685,000 in 2020 to 1 million. The most dramatic increases will occur in low Human Development Index (HDI) countries, where both new cases and deaths are set to double (from 110,000 to 216,000 cases and 59,000 to 116,000 deaths), increasing their proportion of global cases from 18.4% to 22.2% and deaths from 30.1% to 35.2%, mainly as a result of population growth and aging [2]. The drawbacks and limitations of the traditional breast imaging approaches inspired researchers to create and develop novel microwave-based techniques [3].

Recent studies in antenna design for breast tumor detection have focused on scattering parameter (S11) variations to differentiate cancerous from healthy tissues [4–11]. Ganesh et al. [4] offered a wearable monopole array antenna measuring $17 \times 34 \times 0.508$ mm³ that operates at 3.2-4.2 GHz and achieved high reflection co-efficient of -20 dB. In contrast F. A. Ritu et al. [5] proposed a patch antenna performed significantly lower reflection coefficient of -56.204 dB shifts at 2.404 GHz. In [6] a low loss material used and achieved high gain about 7.64 dB in frequency range 7-7.6 GHz but having good gain their proposed antenna [6, 7] showed low penetration characteristics in high frequency region. In [7] P. Yashaswini et al. evaluated its effectiveness using DAS (Delay and Sum) algorithm for imaging localized tumor position. A. Bousrout et al. [8], introduced an ISM-band antenna with very SAR value of 0.283 W/kg, where Kumar et al. [9] fabricated a 90×40 mm² dual-port polarized antenna with a semi-circular slot achieving SAR of 0.179 W/kg at 2.45 GHz. In [10] textile-based antenna fabricated by Z. Ali et al. where its SAR performance are compared by varying the separation distance from the skin surface and limited in FCC threshold value (<1.6 W/Kg) and lastly Gour et al. [11] designed a dual-band wearable antenna with a return loss of -35 dB and -24 dB at 2.43 GHz and 3.32 GHz respectively, which support biomedical applications. A review of studies [4–11] analyzing various antenna designs and their applications in breast tumor detection has been conducted. It is comprehended that microwave antennas face several challenges, including large dimensions [6, 8, 9], low radiation strength [4, 5, 10], limited penetration [6, 7], and the need for improve resolution and sensitivity to minimize false negative in tumor detection.

In this research, a miniaturized compact antenna is designed to address those challenges, and the antenna will make usable, particularly for multilayer breast tumor tissue. The antenna parameters like, return loss, gain, directivity, input impedance with matching characteristics, SAR values etc., tuned into the reconfigurable properties of antenna for high-resolution imaging and deep tissue penetration.

II. METHODOLOGY

The non-invasive tumor-detecting procedure utilizes a biomedical microwave antenna positioned above a breast model at certain distance or on a body surface (Fig. 1). When the antenna sends electromagnetic waves into the layered tissues of the breast model, the interaction between the waves and the layers causes both scattering and reflection. The scattering and reflections occur due to the multi-layer variations in electrical properties, specifically permittivity and conductivity, in different tissue types in the breast model that

impede wave propagation and cause portions of the wave energy to be reflected back towards the antenna while the other wave energy is scattered in different directions. In addition, the electromagnetic characteristics of the tumorous tissues in comparison to healthy tissues, with their high water content and cell densities, produce dielectric properties which alter the behavior of the electromagnetic waves and allow for the identification of tumors based on variations in the received signals. Signals are analyzed using comparisons of the return loss (S_{11}) and other electromagnetic parameters to the typical values for both healthy and tumorous conditions. Signal processing techniques (filtering) determine the existence and localization of tumors based upon the alterations in the received signal. An estimation of the location and existence of the tumors is determined by the 3D electromagnetic simulation visualization of the results of the simulations. Finally, numerical simulations and laboratory measurements verify the accuracy and dependability of the tumor detecting procedure. Although this work verified only by simulation based result analysis.

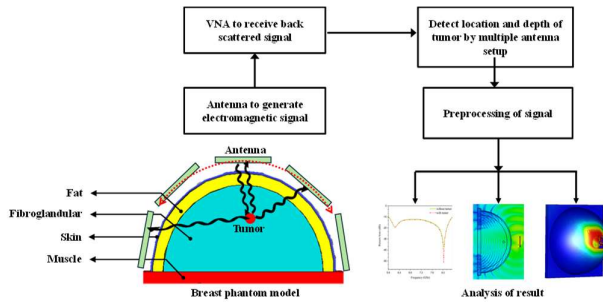


Fig. 1. Outline of the non-invasive detection of breast tumors using microwave technology.

III. DESIGN ANALYSIS

The proposed rectangular microstrip patch antenna is developed in a systematic way with the purpose of optimize in terms of electromagnetic performance. It integrates a FR-4 substrate ($50 \times 50 \times 1.6 \text{ mm}^3$) with a relative permittivity value of 4.4 and a loss tangent value of 0.02. The design of the antenna includes a copper radiator placed on top layer and the continuous copper ground plane underneath. A 1.6 mm wide microstrip feed line is used to ensure good impedance matching with common 50Ω systems. Starting with a simple rectangular patch, square slots are introduced in a strategic way so as to improve bandwidth and create additional resonant modes through the modification of the surface current distribution. Subsequent refinements include the addition of smaller square slots and circular holes for further improving narrowband performance and impedance matching. Ultimately, the patch is very densely patterned with symmetrical square slots of different sizes in order to achieve maximum gain capabilities. The entire ground plane is not changed in the design process, emphasizing the fact that performance improvements result mainly from the changes in patch geometry. Detailed structural views (front, back, cross-sectional, and 3D) are shown in Fig. 3 and key physical dimensions are listed in Table I. All design and performance assessments are conducted with the use of CST simulation software thereby ensuring the validity of the effectiveness of slotting and optimization methodology to enhance the performance of the antenna.

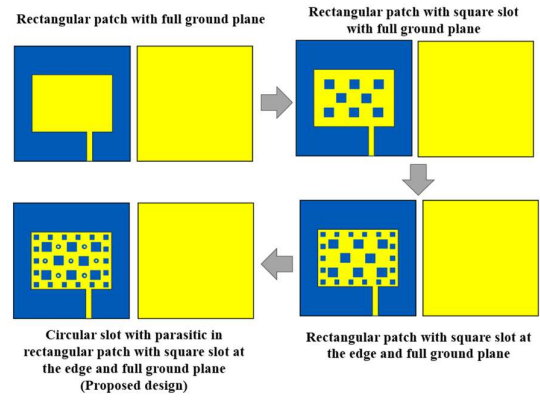


Fig. 2. Design evolution of the proposed slotted patch antenna.

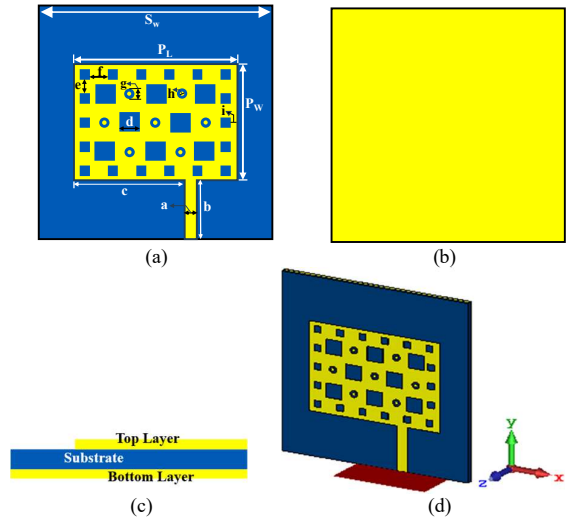


Fig. 3. Detailed view of the proposed antenna design: (a) Front view, (b) Back view, (c) Cross-sectional view, (d) 3D perspective.

TABLE I: PHYSICAL DIMENSION OF THE PROPOSED ANTENNA

Dimension	Unit (mm)	Dimension	Unit (mm)
Substrate length, width (S_L, S_w)	50	Middle square slot (d)	4.2
Patch width (P_W)	35	Distance between two square slots (e, f)	3.4
Patch length (P_L)	25	Circular slot radius (g)	1.0
Feeder width (a)	2.6	Circular parasitic radius (h)	0.5
Feeder length (b)	12.5	Square slot distance from edge (i)	1.5
Left half patch (c)	23.61	Ground length (G_L)	50

IV. PERFORMANCE ANALYSIS

The proposed rectangular slotted patch antenna has several advantageous features to support its use in biomedical applications such as breast tumor detection. It has a low return loss of -41.27 dB at the resonant frequency of 1.9612 GHz shown in Fig. 4. This means the antenna is capable of achieving good matching impedances and thus minimize energy loss while transmitting signals through biomedical imaging devices. Additionally, with a relatively small bandwidth of 0.05 GHz ($1.94\text{--}1.99 \text{ GHz}$) the antenna can have high frequency selectivity which is important for distinguishing between different tissue types and detecting tumors with high specificity.

In Fig. 5, the gain of 4.59 dB at this frequency provides appropriate signal intensity, allowing the antenna to detect minor variations in tissue characteristics during breast tumor

detection. The antenna directivity of 5.635 dBi (Fig. 6) increases power concentration for accurate tumor localization, while the unity VSWR of approximately 1.02 ensures efficient power transfer and high sensitivity. Its impedance of $53.92-j0.88$ Ohms supports compatibility with biomedical equipment.

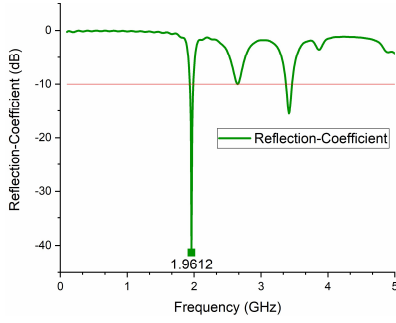


Fig. 4. Reflection co-efficient curve of the proposed patch antenna.

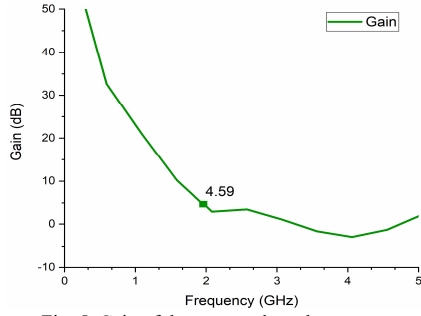


Fig. 5. Gain of the proposed patch antenna.

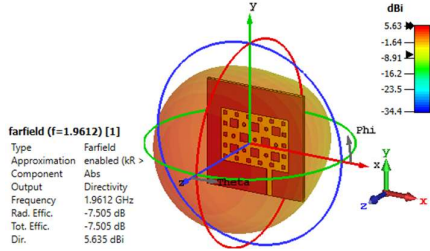


Fig. 6. Radiation pattern (directivity) of the proposed patch antenna.

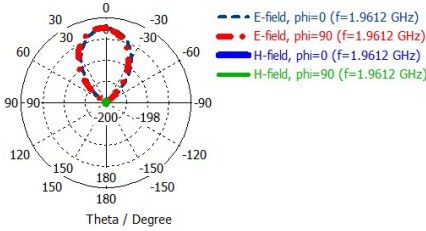


Fig. 7. E, H-field for the proposed patch antenna at $\Phi=0^\circ, 90^\circ$.

Fig. 7 shows the E-field and H-field radiation patterns at 1.9612 GHz for $\phi = 0^\circ$ and $\phi = 90^\circ$, indicating main lobe directions at 0° and 180° with consistent angular widths around $97^\circ-99^\circ$ for E-field. Both the E- and H-field configurations are identical on both planes, with an approximate -196.5 dB (V/m) and approximately -200 dB(A/m), respectively, as well as being very symmetrical and consistent; therefore, we can conclude that both the electric and magnetic fields behave uniformly in all three orientations. As such, it is clear that there will be negligible levels of interference to the signals and, consequently, the signals will have enhanced levels of propagation through body tissues, thus improving the overall

sensitivity of the entire system. The far-field radiation pattern has been achieved with moderate efficiency and directionally focused toward the specified region of the breast tissue for the detection of tumors, thus supporting the suitability of this antenna design for the precise, non-invasive detection of tumors located within the breast. The performance characteristics of a microstrip antenna may vary due to bending, including but not limited to, the resonant frequency, impedance, gain, and radiation pattern when used in wearable or conformal designs. The effect of bending surface curvature on the performance of microstrip antenna is explored in Fig. 8, where bending radius from 0 mm to 250 mm create minimal impact on the resonant frequency and impedance of the antenna that maintains stable performance (Fig. 9).

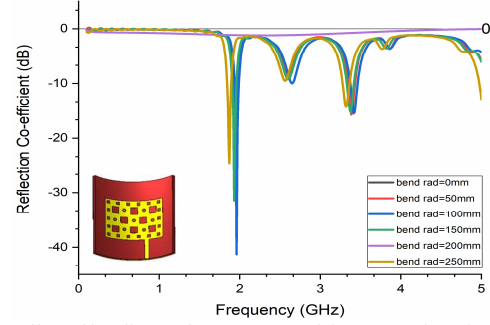


Fig. 8. Effect of bending surface curvature of the proposed patch antenna.

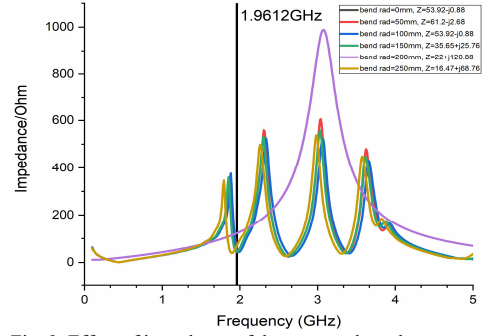


Fig. 9. Effect of impedance of the proposed patch antenna.

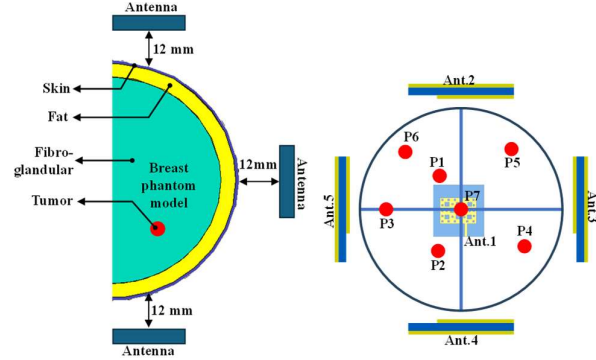


Fig. 10. (a) Cross-section view and (b) 2D view of the place of tumor in breast phantom.

TABLE II. TISSUE PROPERTIES OF BREAST PHANTOM MODEL [12]

Tissues	Permittivity (ϵ)	Conductivity (σ)	Density (ρ)	Thickness (mm)
Skin	38	1.4	1109	2
Fat	5.1	0.1	911	6
Fibroglandular	45	1.5	1041	144
Tumor	55	2	1050	10

V. HUMAN BREAST MODEL AND PROPERTIES

An 80 mm radius 2D cross-sectional human breast phantom model with four concentric layers: skin, fat, fibroglandular tissue, and tumor employed in this proposed work in Fig. 2. This layered phantom model was developed to simulate breast cancer through microwave imaging and each layer represents unique electromagnetic properties such as permittivity and conductivity that are necessary for the simulation of breast cancer detection via microwave imaging. As tumors have greater dielectric constants than normal tissue due to the increased water content, the phantom model will be used to study the differential backscattering patterns between normal tissue and tumor tissue using the values listed in Table II [12].

VI. SENSING PERFORMANCE EVALUATION

There are five antennas surrounding the hemisphere breast phantom (Fig. 11a) positioned for optimal tumor detection from multiple angles. Each antenna connects to a separate port for individual testing. In Fig. 11(b), tumors were placed in different locations of depth to test the accuracy of the detection; it also increased the amount of signal analysis, and increased the reliability of localizing a tumor for advanced image systems.

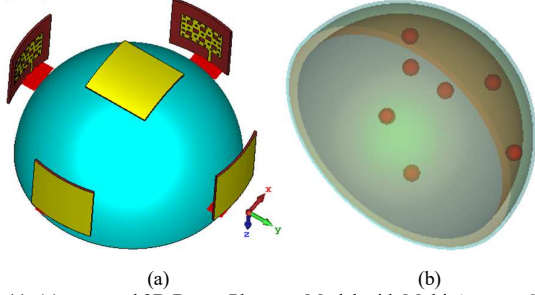


Fig. 11. (a) proposed 3D Breast Phantom Model with Multi-Antenna Setup and (b) 3D view of tumor localization.

When simulating the system, Reflection (S_{11}) and Transmission (S_{21}) coefficient measurements were taken while Antenna-1 transmitted and others received. The significant difference between the two coefficients indicates that there is a tumor present. This method analyzes signal variations to precisely detect and locate breast tumors, offering a non-invasive, accurate breast cancer detection technique.

A. Tumor Localization Analysis

This configuration allows for a detailed analysis of how each antenna interacts with the breast phantom and the tumor inside. The individual roles of the antennas and their respective figures in the simulation are as follows:

1) Antenna-1 as Transmitter (for P1 position):

In this configuration (Fig. 10), antenna-1 is used as the transmitting antenna, while the other four antennas serve as receivers. During this simulation, the transmission coefficient (S_{21}) are measured to evaluate how well the transmitted signal interacts with the tumor and the surrounding tissues in the phantom. Significant changes in the reflection and transmission coefficients are observed in Fig. 12, which provide insight into the location of the tumor.

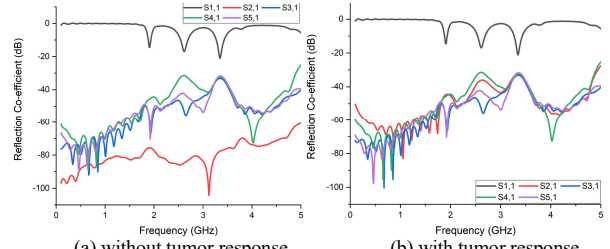


Fig. 12. Signal analysis when Antenna-1 as Transmitter.

2) Antenna-2 as Transmitter (for P1 position):

In this case, antenna-2 is designated as the transmitting antenna (Fig.10), and the remaining antennas receive the signal. The response from the receiving antennas is analyzed to determine the signal variations, particularly the reflection and transmission coefficients. All of these changes in Fig. 13 are indicative of the presence of tumor and provide valuable data for further diagnostic analysis.

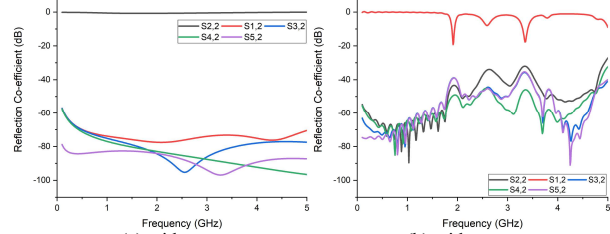


Fig. 13. Signal analysis when Antenna-2 as Transmitter.

3) Antenna-3 as Transmitter (for P1 position):

With Antenna-3 transmitting, antennas 4, 5, 1, and 2 (Fig.10) receive signals. Differences in reflection and transmission coefficients, especially S_{24} (Fig. 14), help accurately locate the tumor within the breast phantom.

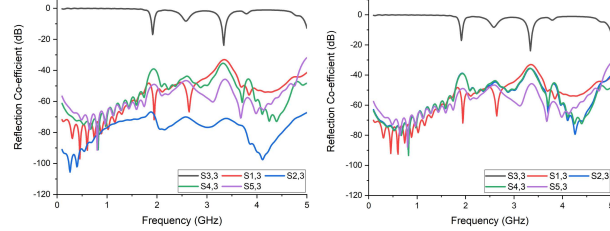


Fig. 14. Signal analysis when Antenna-3 as Transmitter.

4) Antenna-4 as Transmitter (for P1 position):

In Fig. 10, antenna-4 transmits while others receive, showing significant S_{24} variations (Fig. 15). Multiple antenna placements form a complete phantom response map, which improves tumor identification and analysis.

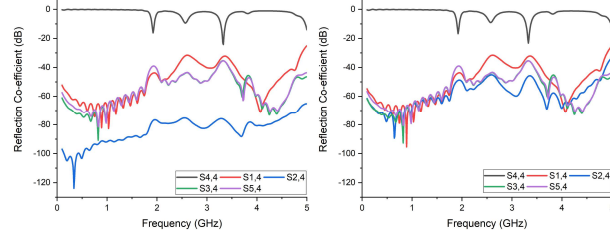
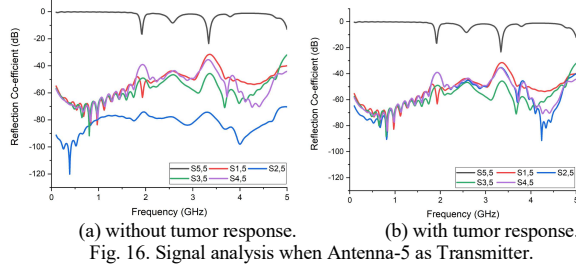


Fig. 15. Signal analysis when Antenna-4 as Transmitter.

5) Antenna-5 as Transmitter (for P1 position):

Finally, in Fig. 16 response of antenna-5 is employed as the transmitting antenna, with the other four antennas serving

as receivers according to Fig. 10 configuration. Again, changes in the transmission coefficients of S15 are measured, helping to predict the tumors presence and location within the breast tissue.



Sequential antenna excitation in a multi-antenna system enhances breast cancer detection by analyzing reflection and transmission coefficient differences to accurately identify tumor presence and location Table III presents changes in reflection coefficients (S-parameters) for different tumor positions (P1–P7) within a breast tissue model when individual antennas are excited. These variations serve as key indicators for tumor detection and location in microwave imaging. For example, excitation of Antenna 1 at Position P1 alters S21, indicating a tumor between Antennas 1 and 2. At Position P2, antenna 2 affects all S-parameters, suggesting a central tumor interacting across the array (Fig. 12-16). Similarly, in antenna 3 excitation impacts S23 and S24, confirming a near-central tumor location (Fig. 17a). Position P3 shows changes only in S52, implying tumor alignment between Antennas 5 and 2 (Fig. 17b). For P4 position in Fig. 17(c), notable changes in S31 and S43 indicate tumor presence near antennas 1, 3, and 4. At position P5, multiple S-parameters (S21, S23, S24) are influenced (Fig. 17d), reflecting a tumor affecting a broader region between antennas 1, 2, 3 and 4. Position P6 in Fig. 17(e) demonstrates changes in S21 and S52, suggesting tumor proximity to antennas 1, 2, and 5. In summary, as shown in Fig. 17(f), position P7 shows a variation across many of the parameters (S21, S42, S32, S43 and S25) that clearly indicates the presence of a tumor near the central axis of the array. The S parameter changes that were observed validate the ability to use this microwave antenna system to detect tumors accurately and determine their location which further supports the application of this technology for early stage breast cancer diagnosis.

TABLE III. ANALYZING REFLECTION COEFFICIENT COMPARED TO HEALTHY CELL

Position	Excitation of Antenna				
	Ant.1	Ant.2	Ant.3	Ant.4	Ant.5
P1	S2,1	All	S2,3	S2,4	S2,5
P2	S2,1	All	S2,3	S2,4	S2,5
P3	NC	S5,2	NC	NC	NC
P4	S3,1	NC	S4,3	NC	NC
P5	S2,1	All	S2,3	S2,4	NC
P6	S2,1	S5,2	NC	NC	NC
P7	S2,1	S4,2	S2,3	S3,4	S2,5

*Ant.=Antenna, NC=No Change

B. Specific Absorption Rate (SAR) Performance

Fig. 18 presents the simulated Specific Absorption Rate (SAR) distribution for the proposed antenna operating at 1.9612 GHz, evaluated using a 1 gram tissue averaging which is significantly below the regulatory limit of 1.6 W/kg for 1 gram of tissue, as specified by the FCC (Federal Communications Commission) guidelines. The 3D distribution indicates localized energy absorption near the

antenna's active region, confirming controlled radiation characteristics. These results demonstrate the antenna's compliance with international safety standards, validating its suitability for safe use in biomedical or wireless communication applications.

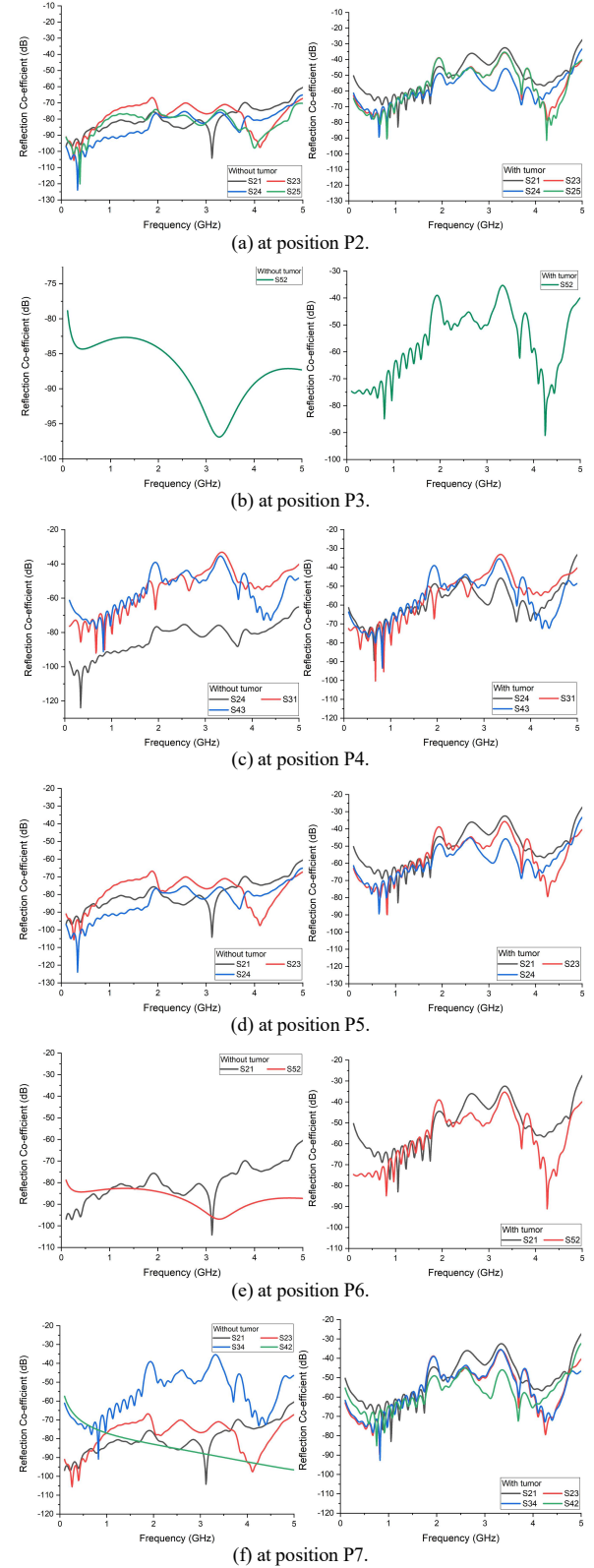


Fig. 17. Reflection co-efficient variation on different positions of tumor location (a-f).

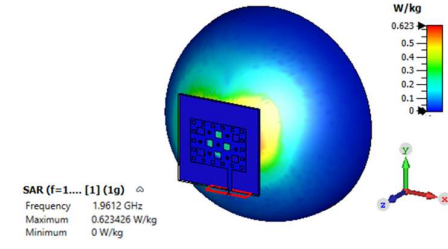


Fig. 18. SAR analysis of proposed design at 1.9612 GHz.

VII. COMPARATIVE ANALYSIS

The Table IV provides a comparative analysis of different wearable antenna designs for breast tumor detection based on parameters like dimensions, operating range, resonance frequency, reflection coefficient, and gain. The proposed work uses an FR-4 substrate and operates within a narrow frequency range (1.94–1.99 GHz). Compared to previous designs, the proposed work has a significantly low profile [5, 6, 8, 9, 10], higher return loss of -41.27 dB [4, 6, 7, 8, 9, 10] which indicates better impedance matching, acceptable gain of 4.59 dB and high penetration capability in lower frequency spectrum suggesting highly effective choice for precise tumor detection.

TABLE IV. PERFORMANCE COMPARISON WITH PREVIOUS RELAVANT WORK

Ref.	Dimension (L×W×H) mm ³	Substrate	Operating range (GHz)	Resonance frequency (GHz)	Reflection Co- efficient (dB)	Gain (dB)
[4]	17×34×0.508	Rogers RT5880	2.2 – 4.2	3.7	~20	–
[5]	50×50×1.5	FR-4	2.36 – 2.45	2.404	-56.204	3.053
[6]	62.475×39.68×1.57	Roger-RT5870	7 – 7.6	7.33	-31.2	7.64
[7]	17.75×15.16×0.51	–	–	8.056	-52.2	7.054
[8]	70×70×1.6	FR-4 Epoxy	2.35 – 2.45	2.4	-32	4.431
[9]	90×40×1.6	FR-4	2.26 – 2.47	2.3	-30	4.63
[10]	50×35×1	Jeans	2 – 3	2.45	-35	–
<i>Proposed Work</i>	<i>50×50×1.6</i>	<i>FR-4</i>	<i>1.94 – 1.99</i>	<i>1.9612</i>	<i>-41.27, -24</i>	<i>4.59</i>

VIII. CONCLUSION

A compact slotted microstrip patch antenna has been designed, simulated, and evaluated to be used as part of a noninvasive breast cancer tumor detector employing microwave imaging. It was found that the proposed antenna operates at a frequency of 1.9612 GHz and is capable of obtaining high performance in terms of return loss, gain, directivity, and low SAR values required in biomedical applications. A multi antenna configuration is employed around a multilayer breast phantom to provide accurate location of tumors via analysis of scattering parameters representing different electromagnetic changes resulting from the presence of tumors. The results show a maximum SAR of 0.623 W/kg, confirming that this antenna complies with the SAR limits established by the FCC and thus can be considered suitable for clinical or wearable application. The antenna also retains consistent performance when moderately bent, which is beneficial for incorporation into flexible or conformable health monitoring devices. In future this simulation based work will be verified by fabricating the proposed antenna and testing through VNA data as well as localized tumor by imaging algorithm. Overall, the proposed microwave-based antenna system demonstrates high sensitivity, safety, and accuracy, offering a promising pathway for early-stage breast cancer diagnosis and future advancements in real-time clinical applications.

REFERENCE

- [1] M. Arnold et al., "Current and future burden of breast cancer: Global statistics for 2020 and 2040," *The Breast*, vol. 66, pp. 15–23, Sep. 2022, doi: 10.1016/j.breast.2022.08.010.
- [2] E. T. Sedeta, B. Jobre, and B. Avezbakiyev, "Breast cancer: Global patterns of incidence, mortality, and trends," *Journal of Clinical Oncology*, vol. 41, no. 16, suppl, p. 10528, Jun. 2023, doi: 10.1200/jco.2023.41.16_suppl.10528.
- [3] S. Iranmakani et al., "A review of various modalities in breast imaging: technical aspects and clinical outcomes," *The Egyptian Journal of Radiology and Nuclear Medicine*, vol. 51, no. 1, Apr. 2020, doi: 10.1186/s43055-020-00175-5.
- [4] T. Ganesh and D. Chaturvedi, "Design of Wearable Monopole Antenna Sensor for Breast Tumor Detection," *2024 IEEE Wireless Antenna and Microwave Symposium (WAMS)*, Visakhapatnam, India, 2024, pp. 1-4, doi: 10.1109/WAMS59642.2024.10527975.
- [5] F. A. Ritu and A. B. M. Aowlad Hossain, "Breast Tumor Detection Using Rectangular Slotted Microstrip Patch Antenna in ISM Band," *2024 IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON)*, Dhaka, Bangladesh, 2024, pp. 55-60, doi: 10.1109/BECITHCON64160.2024.10962745.
- [6] M. Wasif, F. L. Rashid and S. Siddique, "Design and Analysis of an L-Slot Microstrip Patch Antenna for Breast Tumor Detection," *2025 IEEE Region 10 Symposium (TENSYP)*, Christchurch, New Zealand, 2025, pp. 1-7, doi: 10.1109/TENSYP63728.2025.11144951.
- [7] P. Yashaswini, U. Singh and J. Mukherjee, "Enhanced Breast Cancer Detection Using Microwave Imaging with Patch Antenna Arrays and Delay-and-Sum Reconstruction," *2025 3rd IEEE International Conference on Industrial Electronics: Developments & Applications (ICIDEA)*, Bhubaneswar, India, 2025, pp. 1-5, doi: 10.1109/ICIDEA64800.2025.10963192.
- [8] A. Bousrout, A. Khabba, S. Ibnyach, T. Mazri and M. Habibi, "High-Performance Biomedical Antenna for Early and Non-Invasive Breast Cancer Detection," *2025 5th International Conference on Innovative Research in Applied Science, Engineering and Technology (IRASET)*, Fez, Morocco, 2025, pp. 1-5, doi: 10.1109/IRASET64571.2025.11008071.
- [9] H. Kumar and V. V. Thakare, "Design and Development of Dual Port Polarized Microstrip Patch Antenna for Breast Tumor Detection," *2024 IEEE International Conference on Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI)*, Gwalior, India, 2024, pp. 1-6, doi: 10.1109/IATMSI60426.2024.10502759.
- [10] Z. Ali, V. K. Singh, R. Soni, A. Chaudhary, A. Singh and R. Pandey, "Square Patch Antenna for Breast Tumor Detection at 2.45 GHz Frequency," *2025 IEEE 14th International Conference on Communication Systems and Network Technologies (CSNT)*, Bhopal, India, 2025, pp. 995-1000, doi: 10.1109/CSNT64827.2025.10967797.
- [11] S. Gour, A. Rath, and P. Rawal, "A novel dual-band defected ground structure wearable microstrip patch antenna for breast tumor detection in biomedical applications," *Engineering Research Express*, vol. 6, no. 3, p. 035427, Sep. 2024, doi: 10.1088/2631-8695/ad766a.
- [12] C. Gabriel, "Compilation of the dielectric properties of body tissues at RF and microwave frequencies," *Defense Technical Information Center*, Fort Belvoir, VA, 1996.